EXPERIMENT 25: PERFORM SHARPENING OF IMAGE USING GRADIENT MASKING

Aim

To sharpen an image using Gradient Masking with horizontal and vertical gradient operators in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.

2. Read the input image.

3. Convert the image to grayscale.

4. Define the horizontal and vertical gradient masks.

5. Apply the masks to obtain the X and Y gradients.

6. Calculate the gradient magnitude from both gradients.

7. Add the gradient mask to the original image to obtain the sharpened image.

8. Clip the pixel values to the range 0 to 255.

9. Display and save the sharpened image.

Gradient Masks

Horizontal Gradient Mask (Gy):

-1	-2	-1
0	0	0
1	2	1

Vertical Gradient Mask (Gx):

-1	0	1
-2	0	2
-1	0	1

Python Code

```
import cv2
import numpy as np
```

```
# Read the input image
img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Define horizontal gradient mask
gx_kernel = np.array([
    [-1, -2, -1],
    [ 0,  0,  0],
    [ 1,  2,  1]
], dtype=np.float32)

# Define vertical gradient mask
gy_kernel = np.array([
    [-1,  0,  1],
    [-2,  0,  2],
    [-1,  0,  1]
], dtype=np.float32)

# Apply gradient masks
gx = cv2.filter2D(gray, cv2.CV_32F, gx_kernel)
gy = cv2.filter2D(gray, cv2.CV_32F, gy_kernel)

# Calculate gradient magnitude
gradient = np.sqrt(gx ** 2 + gy ** 2)
gradient = np.clip(gradient, 0, 255)

# Perform gradient masking
sharpened = cv2.addWeighted(
    gray.astype(np.float32), 1.0,
    gradient.astype(np.float32), 1.0,
    0
)

# Clip values and convert to 8-bit
```

```
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

# Save output
cv2.imwrite("gradient_masking_output.jpg", sharpened)

# Display images
cv2.imshow("Original Image", gray)
cv2.imshow("Gradient Masking Output", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input

Output

Result

Thus, the image was successfully sharpened using Gradient Masking by enhancing the edges obtained from the horizontal and vertical gradient operators.